

INDIGENE

GROSS ANATOMY OF THE UPPER LIMB

ANA217 — Complete Exam-Ready Compendium

Osteology · Joints & Ligaments · Musculature · Vasculature · Master Hack Sheet

200 Level — Nursing Science, Federal University Oye-Ekiti (FUOYE)
INDIGENE Study Console · Rebuilt for full exam mastery

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1. OSTEOLOGY OF THE UPPER LIMB

The bony skeleton of the upper limb comprises the **pectoral girdle** (scapula + clavicle), the **arm** (humerus), the **forearm** (radius + ulna), and the **hand** (carpals, metacarpals, phalanges). One upper limb has **32 bones total**: clavicle(1) + scapula(1) + humerus(1) + radius(1) + ulna(1) + carpals(8) + metacarpals(5) + phalanges(14).

Clavicle

Alzheimer's Mode: *The clavicle is the only long bone lying **horizontally**, shaped like a stretched-out italic "f" or elongated S — convex forward medially, concave forward laterally. It acts as a strut holding the shoulder out and away from the trunk.*

CORRECTION — OSSIFICATION

The clavicle is the **only bone in the body that ossifies by BOTH mechanisms**: the shaft/body ossifies by **intramembranous ossification** (directly from mesenchyme, no cartilage model), while the **medial (sternal) epiphysis ossifies by endochondral ossification** (via a cartilage model, like a typical long bone) and is notably the **last epiphysis in the entire body to fuse** (~age 25, sometimes later). Do NOT say the clavicle ossifies purely intramembranously — that is a common but incomplete/incorrect statement. The dual mechanism is exactly what makes it unique.

- **Head** (sternal/medial end): rounded, articulates with the manubrium's clavicular notch at the **sternoclavicular joint**.
- **Body/shaft**: transitions from tubular (medial 2/3) to flattened (lateral 1/3); the shape-shift point is marked by the **conoid tubercle** inferiorly.
- **Tail** (acromial/lateral end): flat, blade-like, articulates with the acromion at the **acromioclavicular joint**.
- **Subclavian groove**: inferior surface, middle third — attachment for subclavius muscle.
- **Conoid tubercle** and **trapezoid line**: inferior surface, lateral third — attachment for the coracoclavicular ligament complex (conoid + trapezoid ligaments).

EXAM HACK

Clavicle facts examiners love: (1) first bone to begin ossifying (week 5-6) yet last to finish fusing (~25yo); (2) most commonly fractured bone in the body; (3) fractures classically at the **junction of the middle and lateral thirds** — the weakest point where curvature changes and the cross-section thins; (4) subclavian vessels and brachial plexus lie just deep to the middle third, so severely displaced fractures risk neurovascular injury.

CLINICAL

Falls onto the shoulder (not just onto an outstretched hand) commonly fracture the clavicle. Birth-related clavicle fractures can occur during difficult vaginal deliveries with shoulder dystocia — generally benign and heal well in newborns.

Scapula

Alzheimer's Mode: *A flat triangle glued to the back of the ribcage (overlying ribs 2-7), held on by muscle alone — no direct bony joint to the thorax. Remember it as "3-3-2-3": 3 borders, 3 angles, 2 surfaces, 3 processes.*

Feature	Details
3 Borders	Superior (shortest, sharpest — has suprascapular notch), Medial/vertebral (longest), Lateral/axillary (thickest)
3 Angles	Superior angle (~2nd rib level), Inferior angle (~7th rib level — landmark for counting ribs/vertebral levels), Lateral angle (thickest — bears glenoid cavity)
2 Surfaces	Costal/anterior (concave, subscapular fossa — houses subscapularis), Posterior (divided by the spine into supraspinous and infraspinous fossae)
3 Processes	Spine (continues as acromion), Coracoid process (hook-shaped, palpable below clavicle), Glenoid cavity (shallow socket bordered by supraglenoid tubercle above, infraglenoid tubercle below)

- **Suprascapular notch**: superior border, medial to the coracoid process — transmits the **suprascapular NERVE** only (passes beneath the superior transverse scapular ligament).
- **Greater scapular notch**: superior border, lateral to the coracoid (at the base of the spine) — transmits the **suprascapular ARTERY AND NERVE together** from the supraspinous to the infraspinous fossa.

EXAM HACK

"Suprascapular NERVE gets NOTCHED" — the suprascapular nerve alone passes through the **suprascapular** notch. The artery joins the nerve only at the **greater** scapular notch. Also remember "Army over Navy" — the suprascapular Artery passes OVER the superior transverse scapular ligament while the Nerve passes UNDER it, through the suprascapular notch — a classic entrapment site (suprascapular nerve entrapment syndrome).

CLINICAL

Winging of the scapula results from long thoracic nerve injury (serratus anterior paralysis) — tested with the wall push-up test. Scapular fractures are uncommon and signal high-energy trauma given the bone's mobility and muscular protection.

Humerus

Alzheimer's Mode: *The humerus is "the story of three landmarks that trap three nerves." Learn where each nerve runs relative to the bone and you can answer almost any clinical humerus question.*

Region	Landmark
Proximal	Head (articulates with glenoid) · Anatomical neck · Greater tubercle (SIT muscles insert) · Lesser tubercle (subscapularis inserts) · Intertubercular/bicipital groove (long head of biceps tendon, held by the transverse humeral ligament) · Surgical neck
Shaft	Deltoid tuberosity (lateral, mid-shaft) · Radial/spiral groove (posterior, below deltoid tuberosity)

Distal	Capitulum (lateral, meets radius) · Trochlea (medial, meets ulna) · Medial epicondyle (ulnar nerve groove behind it) · Lateral epicondyle (common extensor origin) · Olecranon fossa (posterior) · Coronoid fossa & radial fossa (anterior)
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EXAM HACK — THE 3-NERVE RULE (own this cold)	
Site	Nerve at risk
Surgical neck	Axillary nerve (+ posterior circumflex humeral artery)
Radial/spiral groove (mid-shaft)	Radial nerve (+ profunda brachii artery)
Medial epicondyle (groove behind it)	Ulnar nerve — the "funny bone" spot
This single table answers a huge share of upper limb clinical questions. Don't mix up surgical neck (axillary) with radial groove (radial) — they are the two most confused sites.	

CLINICAL
Surgical neck fracture → axillary nerve injury → deltoid paralysis + "regimental badge" sensory loss. Mid-shaft fracture → radial nerve injury → wrist drop. Supracondylar fracture (common in children) → risk to brachial artery and median nerve. Lateral epicondylitis = "tennis elbow" (common extensor origin); medial epicondylitis = "golfer's elbow" (common flexor origin).

Radius and Ulna

Alzheimer's Mode: "ROLU, ULR" — Radius is On the Lateral side (thumb side) in anatomical position; Ulna is Medial. The ulna is the fixed pivot; the radius rotates around it like a dial spinning around a fixed post.

Ulna (medial, "stable")	Radius (lateral, "mobile")
Olecranon process (point of elbow, triceps insertion) + coronoid process form the C-shaped trochlear notch	Head (disc-shaped, articulates with capitulum + radial notch of ulna)
Radial notch (lateral side of coronoid process — proximal radioulnar joint)	Radial tuberosity (biceps brachii inserts here)
Ulnar tuberosity (below coronoid — muscle attachment)	Oblique line (descends from tuberosity — muscle attachment)
Supinator crest (lateral shaft — muscle attachment)	Shaft widens progressively distally (opposite of ulna)
Head + styloid process (distal end, small) — does NOT articulate directly with carpal bones (separated by the articular disc/TFCC)	Ulnar notch (distal radioulnar joint) + wide distal end articulating directly with scaphoid + lunate

Both bones are connected proximally (proximal radioulnar joint) and distally (distal radioulnar joint), plus along their shafts by the **interosseous membrane**, which stabilizes the two bones during pronation/supination.

EXAM HACK — FOREARM FRACTURE TRIO
Colles' fracture = distal radius fracture, dorsal displacement (FOOSH injury, elderly) → "dinner fork" deformity.
Smith's fracture = distal radius fracture, volar/anterior displacement (reverse Colles', "garden spade" deformity).
Monteggia fracture = proximal ulna fracture + radial head dislocation ("Monteggia Ulna, proximal").
Galeazzi fracture = distal radius fracture + distal radioulnar joint dislocation ("Galeazzi Radius, distal").
Nightstick fracture = isolated ulnar shaft fracture from a direct blow (classic defensive injury).
Mnemonic: " MUGR " — Monteggia = Ulna (proximal); Galeazzi = Radius (distal).

CLINICAL
The ulna's lack of direct wrist articulation (separated by the fibrocartilaginous disc/TFCC) means the radius transmits the vast majority of force from hand to forearm — this is exactly why distal radius fractures (Colles', Smith's) are so common and clinically significant.

Bones of the Wrist and Hand

Alzheimer's Mode: 8 carpal bones in two rows of 4, like a "roundabout of small stones" between the forearm and the metacarpals.
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Proximal row (lateral → medial): Scaphoid (articulates with radius) · Lunate (articulates with radius) · Triquetrum (articulates with the distal row) · Pisiform (sesamoid bone within flexor carpi ulnaris tendon, sits atop triquetrum — the bulging bone of the wrist).

Distal row (lateral → medial): Trapezium (with its tubercle — articulates with scaphoid + thumb metacarpal, forms the saddle joint) · Trapezoid (articulates with index metacarpal) · Capitate (largest carpal bone, articulates with middle metacarpal) · Hamate (with its hook — articulates with ring + little metacarpals).

EXAM HACK
Mnemonic (proximal row → distal row, lateral → medial): " Some Lovers Try Positions That They Cannot Handle " — Scaphoid, Lunate, Triquetrum, Pisiform, Trapezium, Trapezoid, Capitate, Hamate.
Scaphoid = most commonly fractured carpal bone; blood supply enters distally and runs retrograde , so fractures near the proximal pole risk avascular necrosis (same theme as the femoral neck and talus in the lower limb).
Lunate = most commonly dislocated carpal bone.
Pisiform = smallest carpal, a true sesamoid bone. Capitate = largest carpal bone.

Each metacarpal (5 total, numbered I-V thumb→little finger) has a base, shaft, and head. The 1st metacarpal forms a **saddle joint** with the trapezium, enabling opposition. Phalanges: thumb has 2 (no middle phalanx), fingers 2-5 each have 3 (proximal, middle, distal) — 14 phalanges total.

CLINICAL
Boxer's fracture = neck of the 5th metacarpal (from punching). Bennett's fracture = fracture-dislocation at the base of the 1st metacarpal. Hook of hamate fracture = seen in golfers/racket-sport athletes from repetitive gripping stress. Snuffbox tenderness after a fall → always suspect scaphoid fracture until proven otherwise.

2. JOINTS AND LIGAMENTS OF THE UPPER LIMB

Sternoclavicular (SC) Joint

Alzheimer's Mode: The only bony connection between the entire upper limb and the axial skeleton — a synovial joint reinforced by 4 ligaments, with an articular disc for shock absorption.

- **Anterior sternoclavicular ligament:** clavicular head → anterior clavicular notch of manubrium.
- **Posterior sternoclavicular ligament:** clavicular head → posterior clavicular notch.
- **Costoclavicular ligament:** inferior proximal clavicle → 1st rib + its costal cartilage.
- **Interclavicular ligament:** connects both clavicular heads across the top of the manubrium.

Other Clavicular Ligaments (Coracoclavicular Complex + AC Ligament)

- **Conoid ligament:** conoid tubercle (clavicle) → base of coracoid process.
- **Trapezoid ligament:** trapezoid line (clavicle) → superior surface of coracoid process.
- Conoid + Trapezoid together = the **coracoclavicular ligament** — the primary stabilizer of the distal clavicle/AC joint.
- **Acromioclavicular ligament:** reinforces the AC joint capsule directly (distal clavicle ↔ acromion).

EXAM HACK

AC joint separations are graded by which ligaments tear: Grade I = AC ligament sprain only; Grade III+ = both AC AND coracoclavicular ligaments torn, causing visible step deformity. The coracoclavicular ligament is the TRUE stabilizer — its integrity is what surgeons check on stress X-rays.

Glenohumeral (Shoulder) Joint

Alzheimer's Mode: A golf ball (humeral head) sitting on a golf tee (shallow glenoid). Greatest range of motion of any joint in the body — at the cost of being the most frequently dislocated joint. Stability here is almost entirely soft-tissue, the exact opposite of the hip.

Type: Synovial ball-and-socket joint.

Static Stabilizers

- **Glenoid labrum:** fibrocartilaginous rim, deepens the socket.
- **Joint capsule:** deliberately loose/lax to allow the huge range of motion.
- **Glenohumeral ligaments** (capsule thickenings): Superior GHL (below supraglenoid tubercle → superomedial greater tubercle), Middle GHL (anterior glenoid rim → mid-anatomical neck), Inferior GHL (inferior glenoid rim → inferior surgical neck) — the **IGL is the most important, acting like a hammock, and is the primary restraint against anterior dislocation**.
- **Coracohumeral ligament:** coracoid process → greater + lesser tubercles; supports the weight of the arm.
- **Coracoacromial ligament:** coracoid process → acromion; forms the coracoacromial arch (a protective roof), NOT a true joint ligament (no humeral attachment).
- **Transverse humeral ligament:** bridges the intertubercular groove, holding the long head of biceps tendon in place.

Dynamic Stabilizer

The **rotator cuff** (SITS muscles) actively holds the humeral head centered in the glenoid — the shoulder's stability is muscle-dominant, unlike the hip's ligament-dominant design.

Bursae of the Glenohumeral Joint

Bursa	Location	Note
Subacromial (subdeltoid)	Below acromion/deltoid, above supraspinatus tendon	Most clinically important — subacromial bursitis + impingement occur here together
Subcoracoid	Between coracoid process and middle GHL	Often continuous with subacromial bursa
Subscapular	Below subscapularis tendon, anterosuperior scapula	Usually communicates with the joint cavity directly

EXAM HACK

Static vs Dynamic stabilizers is THE defining shoulder concept examiners test. The coracoacromial arch (coracoacromial ligament + acromion + coracoid process) forms a rigid roof over supraspinatus — repetitive overhead movement pinches the tendon here → subacromial impingement syndrome, often WITH subacromial bursitis since they share the same tight space.

CLINICAL

Anterior shoulder dislocation (~95% of all shoulder dislocations) tears the IGL/anteroinferior labrum → **Bankart lesion**. The humeral head impacts the anterior glenoid rim → **Hill-Sachs lesion** (posterolateral humeral head compression fracture) — the two are commonly found together. Always test sensation over the "regimental badge" area (axillary nerve) after any shoulder dislocation. **Adhesive capsulitis (frozen shoulder):** capsule thickens/contracts, common in diabetics. **Painful arc** (60-120° abduction) = classic sign of subacromial impingement.

Elbow Joint

Alzheimer's Mode: A pure hinge joint — flexion/extension only, no rotation (rotation happens at the separate radioulnar joints, not here).

- **Type:** Synovial hinge joint — humeroulnar (trochlea-trochlear notch) + humeroradial (capitulum-radial head) articulations.

- **Radial collateral ligament:** lateral epicondyle → anterolateral radius (blends with annular ligament).
- **Ulnar collateral ligament:** medial epicondyle → proximal ulna (anterior band is the primary stabilizer — the one reconstructed in "Tommy John surgery").
- **Annular ligament:** wraps around the radial head, anchoring it against the ulna's radial notch — permits pronation/supination while keeping the radial head in place.

CLINICAL

Supracondylar humeral fracture (children) risks the brachial artery and median nerve, which lie just anterior to it. "Pulled elbow" (radial head subluxation) in toddlers occurs when the annular ligament slips over the radial head from a sudden pull on the arm. Ulnar collateral ligament injury is common in throwing athletes.

Wrist (Radiocarpal) Joint

Alzheimer's Mode: Formed between the distal radius and the proximal carpal row (mainly scaphoid + lunate) — the ulna is NOT directly part of this joint, separated by the articular disc.

- **Radial collateral ligament:** radial styloid process → scaphoid + trapezium; prevents excessive adduction (ulnar deviation is limited).
- **Ulnar collateral ligament:** ulnar styloid process → triquetrum + pisiform; prevents excessive abduction (radial deviation is limited).
- **Palmar radiocarpal ligament:** palmar distal radius → carpal bones; aids supination.
- **Dorsal radiocarpal ligament:** dorsal radius → carpal bones/thumb-index metacarpals; aids pronation.
- **Palmar ulnocarpal ligament:** ulnar styloid → lunate + triquetrum.

EXAM HACK

Don't confuse the naming direction with the movement it resists — radial collateral ligament resists ulnar deviation (adduction of the hand), NOT radial deviation. It's named for its location, not the movement it allows.

Carpometacarpal (CMC) and Metacarpophalangeal (MCP) Joints

- **CMC joints:** distal carpal row ↔ metacarpal bases. The 1st CMC (thumb) is a unique saddle joint, allowing opposition; the others are largely plane/gliding joints.
- **MCP joints** (the "knuckles"): condyloid joints, reinforced by the **circumferential volar (palmar) plate, collateral ligaments** on each side, and **3 deep transverse metacarpal ligaments** (linking the index, middle, and ring MCP joints together, limiting their spread).

Interphalangeal (IP) Joints

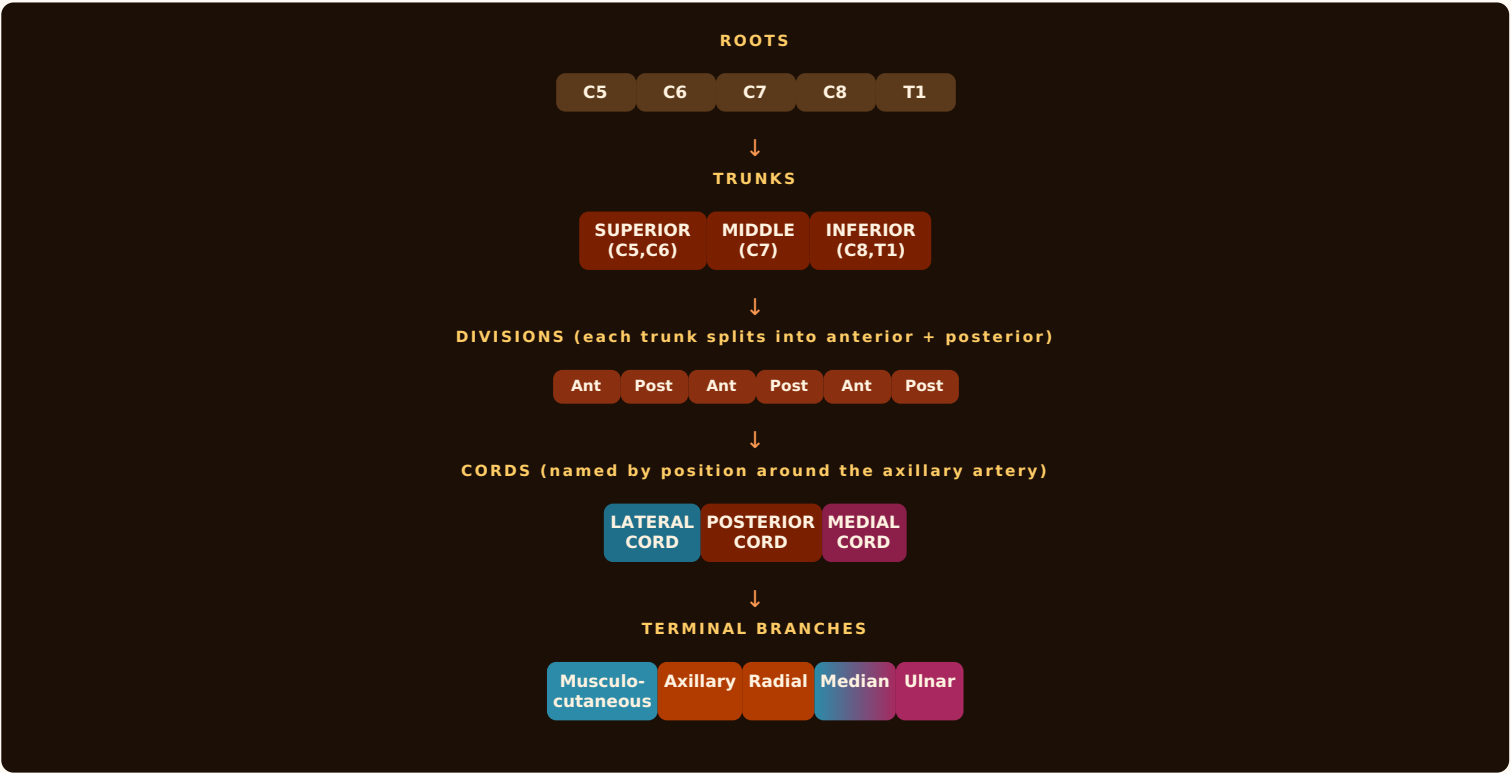
Pure hinge joints (flexion/extension only) between the phalanges — PIP and DIP in fingers 2-5; the thumb has only a single IP joint (no middle phalanx).

EXAM HACK — JOINT TYPES SUMMARY

The **glenohumeral joint** is the ONLY true ball-and-socket joint in the upper limb. The **elbow and interphalangeal joints** are both pure hinge joints. The **1st CMC (thumb)** is a unique saddle joint. Everything else (SC, AC, radioulnar, most CMC, MCP) falls somewhere between plane and condyloid — but glenohumeral = ball-and-socket and elbow/IP = hinge are the two facts examiners test most.

3. NERVE SUPPLY — THE BRACHIAL PLEXUS

Alzheimer's Mode: Picture a highway system — 5 roots merge into 3 trunks, each trunk splits into 2 divisions, and the divisions regroup into 3 cords, which finally branch into 5 terminal nerves. Remember it as **"Randy Travis Drinks Cold Beer"** — Roots, Trunks, Divisions, Cords, Branches.



Cord	Formed from	Gives rise to
Lateral cord	Anterior divisions of superior + middle trunks (C5,6,7)	Musculocutaneous nerve (alone) + lateral head of median nerve
Posterior cord	Posterior divisions of ALL THREE trunks (C5-T1) — the only cord touched by every root	Axillary nerve + Radial nerve
Medial cord	Anterior division of inferior trunk only (C8,T1)	Ulnar nerve (alone) + medial head of median nerve

EXAM HACK
Mnemonic for the 5 terminal branches: **"Matt Ate Rats And Mice"** — Musculocutaneous, Axillary, Radial, (median straddles both), Ulnar. The **median nerve is the only nerve formed from TWO cords** (lateral + medial), giving it its characteristic "M" shape straddling the axillary artery. **Posterior cord = only cord touching every root (C5-T1)** because it collects the posterior division of all 3 trunks — this is why posterior cord injury has the broadest deficit of the three cords.

EXAM HACK — PRE-TRUNK (ROOT-LEVEL) BRANCHES
Some nerves branch directly off the ROOTS, before the trunks even form — a classic trick question lists these among "trunk branches" when they're not. Includes: **Long thoracic nerve** (C5,6,7 → serratus anterior), **Dorsal scapular nerve** (C5 → rhomboids + levator scapulae), **Nerve to subclavius** (C5,C6 → subclavius). Also root-level: **Suprascapular nerve** branches off the **superior trunk** (not a root) → supraspinatus + infraspinatus.

CLINICAL — ERB'S vs KLUMPKE'S PALSY
Erb's palsy (upper trunk, C5-C6): birth trauma or motorcycle accident forcing the head away from the shoulder → **"waiter's tip" position** (arm adducted, medially rotated, elbow extended, forearm pronated, wrist flexed) — deltoid, supraspinatus, infraspinatus, biceps, brachialis all weak. "Erb's = Elevated/upper trunk = can't lift the arm."
Klumpke's palsy (lower trunk, C8-T1): arm pulled upward forcefully (breech delivery, grabbing something while falling) → **claw hand** (loss of intrinsic hand muscles) ± **Horner's syndrome** if T1 root is avulsed (ptosis, miosis, anhidrosis — sympathetic fibers travel with T1). "Klumpke's = can't use the hand."
Long thoracic nerve injury → winged scapula (serratus anterior palsy). **Winging tested with the wall push-up test.**

4. MUSCULATURE OF THE UPPER LIMB

Upper limb muscles are grouped by region: scapular stabilizers (back), shoulder-joint movers (rotator cuff + deltoid + teres major), thoracic wall muscles, the arm, the forearm, and the intrinsic hand muscles.

Muscles Arising from the Back (Scapular Stabilizers)

Alzheimer's Mode: These are the "puppet strings" of the scapula — since the scapula has no direct bony joint to the ribcage, these muscles anchor and reposition it, giving the rotator cuff a stable base to work from.

Muscle	Origin	Insertion	Action	Nerve
Trapezius	Superior nuchal line, occipital protuberance, nuchal ligament, spinous processes C7-T12	Lateral clavicle, acromion, scapular spine	Elevates (upper), retracts (middle), depresses (lower) scapula; all parts rotate scapula upward	Accessory nerve (CN XI) + C3,C4 proprioceptive
Levator scapulae	Posterior tubercles, transverse processes C1-C4	Superior angle → medial border of scapula	Elevates scapula	C3, C4 + dorsal scapular nerve
Rhomboid minor	Nuchal ligament + spinous processes C7-T1	Medial border, at level of scapular spine	Retracts, stabilizes, assists elevation/rotation	Dorsal scapular nerve
Rhomboid major	Spinous processes T2-T5	Medial border, below the spine (below rhomboid minor)	Retracts, stabilizes scapula	Dorsal scapular nerve
Serratus anterior	Outer surfaces of ribs 1-8/9	Costal (anterior) surface of the medial border of scapula	Protracts scapula; rotates it upward for full overhead abduction	Long thoracic nerve
Latissimus dorsi	Spinous processes T6-L2, ribs 8-12, thoracolumbar fascia, iliac crest	Floor of intertubercular groove of humerus	Adduction, extension, internal rotation of the arm	Thoracodorsal nerve

EXAM HACK

Trapezius is the **ONLY** muscle here supplied by a **cranial nerve** (CN XI) rather than a plexus branch — a classic "odd one out" trap. Levator scapulae + both rhomboids share the same **dorsal scapular nerve** (a pre-trunk root-level branch of the brachial plexus, arising directly from C5, before the trunks even form — same pattern as the long thoracic nerve). Rhomboid minor sits **ABOVE** rhomboid major (minor = C7-T1, major = T2-T5, lower). **Serratus anterior + trapezius = the force couple for full overhead arm raising** — if either is weak, full abduction fails even with a perfect rotator cuff. Latissimus dorsi's insertion completes a 3-muscle trio at the intertubercular groove: **pectoralis major (lateral lip), latissimus dorsi (floor), teres major (medial lip)** — "a Lady between two Majors."

CLINICAL

Long thoracic nerve injury → winged scapula (serratus anterior palsy), tested with the wall push-up test. Accessory nerve injury (e.g., neck lymph node biopsy) → trapezius weakness, shoulder droop, difficulty raising the arm above horizontal. Dorsal scapular nerve injury → subtle winging + weak retraction. Latissimus dorsi flap is used surgically in breast reconstruction; thoracodorsal nerve injury (e.g., during axillary lymph node dissection) weakens climbing/pulling movements.

Muscles Arising from the Scapula (Rotator Cuff + Shoulder Movers)

Alzheimer's Mode: The rotator cuff is 4 muscles acting like guy-wires holding the humeral head steady in the shallow glenoid "saucer" — without them, every arm-raise would risk dislocation.

Muscle	Origin	Insertion	Action	Nerve
Supraspinatus (SITS)	Supraspinous fossa	Greater tubercle (top facet)	Initiates first 15° of abduction	Suprascapular nerve
Infraspinatus (SITS)	Infraspinous fossa	Greater tubercle (middle facet)	Lateral (external) rotation	Suprascapular nerve
Teres minor (SITS)	Lateral border of scapula	Greater tubercle (lower facet)	Lateral rotation	Axillary nerve
Subscapularis (SITS)	Subscapular fossa	Lesser tubercle	Medial (internal) rotation	Upper + lower subscapular nerves
Deltoid	Scapular spine, acromion, lateral clavicle	Deltoid tuberosity	Abduction (15-90°, middle fibers), flexion (anterior fibers), extension (posterior fibers)	Axillary nerve
Teres major	Inferior angle of scapula (dorsal surface)	Medial lip of intertubercular groove	Extension, adduction, internal rotation	Lower subscapular nerve

EXAM HACK — SITS = ROTATOR CUFF

Supraspinatus, Infraspinatus, Teres minor, Subscapularis. Three of the four (S, I, T-minor) insert on the **greater tubercle** and — except teres minor (axillary nerve) — are **suprascapular nerve**. Subscapularis is the odd one out: different tubercle (lesser), different action (medial not lateral rotation). **Abduction relay:** 0-15° = supraspinatus → 15-90° = deltoid → beyond 90° = trapezius + serratus anterior rotating the scapula (scapulohumeral rhythm, ~2:1 glenohumeral:scapulothoracic ratio). Teres major is **NOT** part of the rotator cuff despite the similar name to teres minor — easy trap.

CLINICAL

Supraspinatus is the most commonly torn rotator cuff tendon (repetitive overhead strain, impingement under the coracoacromial arch). Axillary nerve injury (surgical neck fracture, anterior dislocation) → deltoid + teres minor paralysis, flattened "squared-off" shoulder, regimental badge sensory loss.

Muscles Arising from the Thoracic Wall

Muscle	Origin	Insertion	Action	Nerve
Pectoralis major	Clavicular head: medial clavicle. Sternocostal head: sternum + costal cartilages ribs 1-6	Lateral lip of intertubercular groove	Adduction, flexion, internal rotation of the arm (glenohumeral joint)	Medial + lateral pectoral nerves (C5-T1)
Pectoralis minor	Anterior surface, ribs 3-5	Coracoid process	Depresses/protracts the scapula ; elevates ribs in forced inspiration	Medial pectoral nerve (C8,T1)
Subclavius	1st rib + costal cartilage	Subclavian groove of clavicle	Depresses the clavicle (and thus the shoulder); cushions the neurovascular bundle beneath the clavicle	Nerve to subclavius (C5,C6)

CORRECTION — DON'T LUMP THESE THREE TOGETHER

Pectoralis major acts on the ARM at the glenohumeral joint — it does NOT move the scapula. **Pectoralis minor and subclavius act on the SCAPULA/CLAVICLE**, not the arm. This is a classic exam trap: examiners deliberately group these three by region (thoracic wall) to test whether you know they don't share the same functional target. Pec major belongs functionally with latissimus dorsi and teres major (all move the humerus); pec minor belongs with the scapular stabilizers (trapezius, rhomboids, levator scapulae, serratus anterior).

EXAM HACK

Pec major's two heads have opposite-feeling actions depending on arm position: the clavicular head flexes the arm (front raise); the sternocostal head extends the arm from a flexed position (pulling something down) — a "why does one muscle have two different actions" trap. Pectoralis minor's position defines the **3 parts of the axillary artery** and the **3 axillary lymph node levels** — the single most repeated landmark in upper limb anatomy. Subclavius physically cushions the subclavian vessels/brachial plexus, which is partly why clavicle fractures don't injure them as often as proximity would suggest.

CLINICAL

Pectoralis major tear/rupture occurs during heavy bench pressing — sudden pain, bruising, weak adduction/internal rotation. Poland syndrome = congenital absence of the sternocostal head of pec major (classic exam mention). Pectoralis minor syndrome = neurovascular compression beneath a tight pec minor, a thoracic-outlet-type presentation.

Scapular Movements — The Full System

The scapula moves in 6 directions, and virtually every muscle you've just learned contributes to at least one:

Movement	Muscles
Elevation (shrugging)	Trapezius (upper fibers) + Levator scapulae
Depression	Trapezius (lower fibers) + Pectoralis minor + Subclavius (indirect)
Protraction (arm adducted, reaching forward)	Serratus anterior + Pectoralis minor
Retraction (arm abducted)	Trapezius (middle fibers) + Rhomboids
Upward rotation (arms raised overhead)	Trapezius + Serratus anterior (force couple)
Downward rotation (arms by side)	Rhomboids + Levator scapulae + Pectoralis minor

Muscles of the Arm — Anterior Compartment

Muscle	Origin	Insertion	Action	Nerve
Biceps brachii	Long head: supraglenoid tubercle. Short head: coracoid process tip	Radial tuberosity + bicipital aponeurosis	Flexes elbow; powerful supinator (strongest at 90° flexion)	Musculocutaneous nerve
Coracobrachialis	Coracoid process tip	Medial humeral shaft	Flexes + adducts the arm at the shoulder	Musculocutaneous nerve
Brachialis	Distal anterior humeral shaft	Ulnar tuberosity	Main flexor of the elbow — works in ALL forearm positions	Musculocutaneous nerve AND radial nerve (dual innervation)

Muscle of the Arm — Posterior Compartment

Muscle	Origin	Insertion	Action	Nerve
Triceps brachii	Long head: infraglenoid tubercle. Lateral head: above the radial groove. Medial head: below the radial groove	Olecranon process	Extends elbow + arm; long head also extends/adducts shoulder	Radial nerve
Anconeus	Lateral epicondyle	Olecranon process	Assists elbow extension; stabilizes ulna during pronation	Radial nerve

EXAM HACK

Brachialis is the true prime elbow flexor — not biceps. Biceps is actually a better SUPINATOR than flexor, and only flexes efficiently when the forearm is supinated. Brachialis flexes in every position. **Brachialis is the only anterior-arm muscle with dual innervation** (musculocutaneous + radial) — a classic "odd one out" question, and a pattern that repeats later with flexor digitorum profundus in the forearm. The radial nerve runs in the spiral groove sandwiched between triceps' lateral and medial heads — this is why mid-shaft humeral fractures classically injure the radial nerve.

CLINICAL

Distal biceps tendon rupture — sudden pop with a "Popeye deformity" bulge in the arm, from forceful eccentric loading (e.g., catching a heavy falling object). Triceps reflex tests the C7 nerve root and radial nerve integrity.

Muscles of the Forearm — Anterior (Flexor) Compartment

Alzheimer's Mode: Almost all originate from ONE spot — the common flexor origin on the medial epicondyle — like ropes tied to a single anchor point. Nerve rule: median nerve runs the show, except ulnar nerve gets exactly 1½ muscles.

Superficial layer

Muscle	Insertion	Action	Nerve
Pronator teres (most lateral)	Lateral radial shaft	Pronation, assists elbow flexion	Median
Flexor carpi radialis	Base of 2nd/3rd metacarpals	Flexion + abduction of wrist	Median
Palmaris longus (variably absent)	Palmar aponeurosis	Flexion of wrist	Median
Flexor carpi ulnaris (most medial)	Pisiform, hamate, 5th metacarpal base	Flexion + adduction of wrist	Ulnar

Intermediate layer

Muscle	Insertion	Action	Nerve
Flexor digitorum superficialis	Middle phalanges of fingers 2-5	Flexes PIP joints (+ wrist)	Median

Deep layer

Muscle	Insertion	Action	Nerve
Flexor digitorum profundus	Distal phalanges of fingers 2-5	Flexes DIP joints	Ulnar (medial half) + Anterior interosseous nerve (lateral half) — split/dual
Flexor pollicis longus	Distal phalanx of thumb	Flexes thumb IP joint	Anterior interosseous nerve (median branch)
Pronator quadratus (deepest)	Distal radius	Pronation	Anterior interosseous nerve

EXAM HACK — THE "1½ MUSCLE" ULNAR RULE

In the entire anterior forearm, the ulnar nerve supplies ONLY: flexor carpi ulnaris (1 full muscle) + the medial half of flexor digitorum profundus (½ muscle). Everything else is median territory. **FDP mirrors brachialis's dual-innervation trap** — both are the "exception" muscle in their compartment. To test FDS alone clinically: hold the other 3 fingers straight (disables FDP's shared tendon action) and ask the patient to flex the tested finger — if only the PIP flexes, FDS is intact.

CLINICAL

Carpal tunnel syndrome compresses the median nerve at the wrist — but NOT its territory in the forearm (median nerve branches to forearm flexors leave before the carpal tunnel). Anterior interosseous nerve syndrome selectively weakens FPL, lateral FDP, and pronator quadratus — presents as inability to make an "OK sign" (pinch grip).

Muscles of the Forearm — Posterior (Extensor) Compartment

Alzheimer's Mode: Posterior forearm = radial nerve territory, no exceptions, no split-innervation traps — the "easy half" compared to the flexor side.

Superficial layer (common extensor origin: lateral epicondyle)

Muscle	Insertion	Action	Nerve
Brachioradialis (lateral-most)	Radial styloid process	Flexes the elbow in mid-pronation (the "imposter" of this compartment)	Radial
Extensor carpi radialis longus	Base of index metacarpal	Extension + abduction of wrist	Radial
Extensor carpi radialis brevis	Base of middle metacarpal	Extension of wrist	Posterior interosseous nerve
Extensor digitorum	Middle phalanges of fingers 2-5	Extension of hand + fingers	Posterior interosseous nerve
Extensor digiti minimi	Distal phalanx of little finger	Extension of little finger	Posterior interosseous nerve
Extensor carpi ulnaris (most medial)	Base of 5th metacarpal	Extension + adduction of wrist	Posterior interosseous nerve
Anconeus	Olecranon	Assists elbow extension	Radial nerve

Deep layer

Muscle	Insertion	Action	Nerve
Supinator	Wraps around proximal radius	Supination	Posterior interosseous nerve
Abductor pollicis longus	Base of thumb metacarpal	Abduction/extension of thumb	Posterior interosseous nerve
Extensor pollicis brevis	Proximal phalanx of thumb	Extension of thumb	Posterior interosseous nerve
Extensor pollicis longus	Distal phalanx of thumb	Extension of thumb	Posterior interosseous nerve
Extensor indicis	Proximal phalanx of index	Independent extension of index finger	Posterior interosseous nerve

EXAM HACK

Brachioradialis is the imposter — lives in the extensor/radial nerve compartment but its ACTION is elbow flexion, strongest in mid-pronation (like lifting a mug). Every exam tests this contradiction. **Anatomical snuffbox borders:** lateral = Abductor pollicis longus + Extensor pollicis brevis tendons; medial = Extensor pollicis longus tendon; floor = scaphoid (snuffbox tenderness → suspect scaphoid fracture); the radial artery runs through the floor. **Lateral epicondylitis ("tennis elbow")** = inflammation at the common extensor origin, most often extensor carpi radialis brevis specifically.

Muscles of the Hand — Intrinsic Muscles

Alzheimer's Mode: 4 groups — Thenar (thumb pad), Hypothenar (little finger pad), Central (lumbricals + interossei), and Adductor pollicis (a group of one). Nerve rule: ulnar nerve owns almost everything; median nerve gets only a small VIP section.

Thenar muscles (all median nerve, recurrent branch) — mnemonic "AFO"

Muscle	Action
Abductor pollicis brevis	Abducts the thumb
Flexor pollicis brevis	Flexes the thumb
Opponens pollicis	Opposition (medial rotation) of the thumb

Hypothenar muscles (all ulnar nerve, deep branch) — same naming pattern, mirrored nerve

Muscle	Action
Abductor digiti minimi	Abducts the little finger
Flexor digiti minimi brevis	Flexes the little finger
Opponens digiti minimi	Opposition (lateral rotation) of the little finger

Single/central intrinsic muscles

Muscle	Action	Nerve
Adductor pollicis	Adducts the thumb	Ulnar (deep branch) — the odd one out among thumb intrinsic
Palmar interossei (4)	ADduct fingers toward the middle finger axis	Ulnar (deep branch)
Dorsal interossei (4)	ABduct fingers away from the middle finger axis	Ulnar (deep branch)
Lumbricals (4)	Flex MCP joints while extending IP joints simultaneously	Split: lateral 2 (index/middle) = Median; medial 2 (ring/little) = Ulnar

EXAM HACK

Thenar/hypothenar mirror trick: muscle NAMES mirror exactly (Abductor-Flexor-Opponens on both sides) but the NERVE flips — thumb side = median, little finger side = ulnar. **"PAD and DAB"** — Palmar interossei ADduct, Dorsal interossei ABduct. Lumbricals are the ONLY intrinsic hand muscles with split median/ulnar innervation — mirroring the same split-innervation pattern you already know from brachialis (arm) and flexor digitorum profundus (forearm) — a recurring exam theme across the whole limb. Adductor pollicis is ulnar-supplied despite acting on the thumb — the classic "odd one out" among thumb muscles (all others thenar = median).

CLINICAL

Froment's sign (ulnar nerve palsy test): gripping paper between thumb and index finger — with weak adductor pollicis (ulnar), the patient compensates using flexor pollicis longus (median), flexing the thumb IP joint. **Ulnar claw hand:** hyperextension at MCP + flexion at IP of ring/little finger (loss of lumbricals 3&4). **Median claw/"ape hand"** (carpal tunnel syndrome, severe): thenar wasting, thumb can't oppose.

5. VASCULATURE OF THE UPPER LIMB

The blood supply begins as the subclavian artery (direct branch of the aortic arch on the left; from the brachiocephalic trunk on the right), continuing as the axillary artery, then the brachial artery, bifurcating into the ulnar and radial arteries.

Subclavian Artery and Branches

Branch	Supplies
Vertebral artery (1st branch)	Ascends through cervical transverse foramina → brain
Thyrocervical trunk (2nd branch) — 4 sub-branches	Suprascapular a., Transverse cervical a. (superficial branch + deep branch = dorsal scapular a.), Inferior thyroid a., Ascending cervical a.
Internal thoracic artery	Descends to supply the chest wall
Costocervical trunk (last branch)	Deep cervical a. + Superior intercostal a. (1st two posterior intercostal arteries)

Axillary Artery and Branches

Alzheimer's Mode: Described in 3 parts relative to pectoralis minor, and the branch count matches the part number: "1-2-3."

Part	Branches
1st part (above pec minor) — 1 branch	Superior thoracic artery
2nd part (posterior to pec minor) — 2 branches	Thoracoacromial trunk (4 sub-branches: pectoral, acromial, clavicular, deltoid) + Lateral thoracic artery
3rd part (below pec minor) — 3 branches	Subscapular artery (→ circumflex scapular + thoracodorsal), Anterior circumflex humeral artery, Posterior circumflex humeral artery (larger diameter than anterior)

EXAM HACK

6 branches total, mnemonic: **"Screw The Lawyer, Save A Patient"** — Superior Thoracic, Thoracoacromial, Lateral thoracic, Subscapular, Anterior circumflex humeral, Posterior circumflex humeral. The posterior circumflex humeral artery travels WITH the axillary nerve through the quadrangular space — paired structures, paired injury risk.

Brachial Artery and Branches

The axillary artery becomes the brachial artery at the lower margin of teres major; it terminates at the radial notch of the ulna by bifurcating into the ulnar and radial arteries.

- **Profunda brachii artery:** runs posteriorly through the triangular interval, alongside the radial nerve in the radial/spiral groove; gives the radial collateral artery.
- **Superior ulnar collateral artery:** descends posterior to the medial epicondyle → anastomoses with the posterior recurrent ulnar artery.
- **Inferior ulnar collateral artery:** anastomoses posteriorly with the radial collateral artery and anteriorly with the anterior recurrent ulnar artery.

EXAM HACK

The elbow anastomotic network links: profunda brachii's radial collateral, superior + inferior ulnar collateral arteries, and the anterior + posterior recurrent ulnar arteries. This collateral network matters clinically if the brachial artery is occluded or ligated near the elbow — collateral flow can maintain forearm perfusion.

Ulnar Artery and Branches

- **Anterior ulnar recurrent artery:** ascends to anastomose with the inferior ulnar collateral artery.
- **Posterior ulnar recurrent artery:** ascends posterior to the medial epicondyle → anastomoses with the superior ulnar collateral artery.
- **Common interosseous artery:** splits into anterior interosseous artery (anterior forearm) + posterior interosseous artery (posterior forearm).
- **Volar carpal + dorsal carpal branches** at the wrist.

Radial Artery and Branches

- **Radial recurrent artery:** ascends to anastomose with profunda brachii.
- **Palmar carpal artery:** anastomoses with the ulnar artery's volar carpal branch.

Arteries of the Hand

Alzheimer's Mode: Two palmar arches with opposite dominance, plus a dorsal arch — "Superficial = Ulnar-dominant, Deep = Radial-dominant."

Arch	Formed by	Supplies
Superficial palmar arch	Ulnar artery (main) + superficial branch of radial artery (completes it)	Digital arteries → fingers

Deep palmar arch	Radial artery (main) + deep branch of ulnar artery (completes it)	Deep palm muscles; gives metacarpal branches anastomosing with the superficial arch's digitals
Princeps pollicis artery	Deep palmar branch of radial artery	Thumb (both sides)
Radialis indicis artery	Deep radial palmar artery (before forming deep arch)	Lateral side of index finger
Dorsal carpal arch	Dorsal volar artery + dorsal radial carpal artery	Digital branches on the dorsum of the hand

EXAM HACK

Don't confuse the two palmar arches' dominance: **Superficial = ULNAR-dominant; Deep = RADIAL-dominant**. The **dorsal carpal arch** is a separate third arch on the back of the wrist — don't lump it with the two palmar arches. **Allen's test** checks patency of both radial and ulnar arteries plus the completeness of the palmar arch anastomosis — occlude both, release one at a time, watch color return.

6. MASTER HACK SHEET — LAST-MINUTE REVIEW

Every mnemonic, trap, and high-yield rule from this compendium, compressed onto one section. Use this the night before the exam.

OSTEOLOGY

Clavicle: dual ossification (intramembranous shaft + endochondral medial epiphysis, last to fuse ~25yo); most fractured bone; breaks at middle-lateral third junction.

Scapula: "3-3-2-3" (borders-angles-surfaces-processes); suprascapular NOTCH = nerve only, GREATER notch = artery+nerve.

Humerus 3-nerve rule: surgical neck→axillary; radial groove→radial; medial epicondyle→ulnar.

Ulna=fixed pivot, Radius=rotates. MUGR: Monteggia=Ulna(proximal); Galeazzi=Radius(distal).

Carpals: "Some Lovers Try Positions That They Cannot Handle." Scaphoid=most fractured, retrograde supply→AVN risk. Lunate=most dislocated.

JOINTS & LIGAMENTS

Coracoclavicular ligament (conoid+trapezoid) = TRUE AC joint stabilizer.

Glenohumeral: IGL=primary anterior-dislocation restraint (hammock); coracoacromial ligament ≠ true joint ligament (no humeral attachment); rotator cuff=dynamic, labrum/capsule/GHLs=static.

Subacromial bursa = most clinically important, shares space with supraspinatus (impingement).

Elbow & IP joints = pure hinge. Glenohumeral = ONLY ball-and-socket in upper limb. 1st CMC = saddle joint.

SCAPULAR/SHOULDER MUSCLES

Trapezius = CN XI (only cranial-nerve-supplied muscle here).

Levator scapulae + both rhomboids = dorsal scapular nerve.

Serratus anterior + trapezius = force couple for overhead abduction.

SITS = rotator cuff; 3 of 4 → greater tubercle + suprascapular n. (except teres minor=axillary n.); subscapularis = odd one out (lesser tubercle, medial rotation).

Abduction relay: 0-15°=supraspinatus, 15-90°=deltoid, >90°=trapezius+serratus anterior.

Pec major moves the ARM; pec minor + subclavius move the SCAPULA/CLAVICLE — don't lump together.

Intertubercular groove trio: pec major (lateral lip), lat dorsi (floor), teres major (medial lip).

BRACHIAL PLEXUS

"Randy Travis Drinks Cold Beer" = Roots→Trunks→Divisions→Cords→Branches.

Trunks: Sup=C5,6; Mid=C7; Inf=C8,T1.

Posterior cord=only cord from every root; gives Axillary+Radial.

Lateral cord alone→Musculocutaneous; Medial cord alone→Ulnar; BOTH→Median.

Pre-trunk branches: Long thoracic (serratus ant.), Dorsal scapular (rhomboids/lev.scap), N. to subclavius.

Erb's=upper trunk=waiter's tip. Klumpke's=lower trunk=claw hand ±Horner's.

ARM & FOREARM MUSCLES

Brachialis = true prime elbow flexor (all positions); biceps = better supinator.

Brachialis = dual nerve (musculocutaneous + radial) — "odd one out" #1.

Anterior forearm = median-dominant; ulnar gets exactly 1½ muscles (FCU + medial ½ FDP).

FDP = dual nerve (ulnar + AIN) — "odd one out" #2, mirrors brachialis.

Posterior forearm = 100% radial nerve, no exceptions.

Brachioradialis = imposter (extensor compartment, but FLEXES elbow).

Snuffbox: APL+EPB lateral border, EPL medial border, scaphoid floor, radial artery runs through it.

HAND INTRINSICS

Thenar (AFO) = median n. Hypothenar (mirror names) = ulnar n.

Adductor pollicis = ulnar (odd one out among thumb muscles).

Lumbricals = split: lateral 2=median, medial 2=ulnar — 3rd "odd one out" split-innervation pattern (after brachialis, FDP).

PAD = Palmar ADduct; DAB = Dorsal ABduct.

Froment's sign = ulnar palsy compensation via median-supplied FPL.

VASCULATURE

Axillary artery "1-2-3" branches per part (split by pec minor). Mnemonic: "Screw The Lawyer, Save A Patient."

Posterior circumflex humeral a. travels with axillary nerve (quadrangular space).

Elbow anastomosis: profunda brachii's radial collateral + sup/inf ulnar collateral + ant/post recurrent ulnar arteries.

Palmar arches: Superficial=ULNAR-dominant; Deep=RADIAL-dominant. Dorsal carpal arch = separate third arch.

Allen's test = checks radial+ulnar patency + arch anastomosis.

RECURRING "ODD ONE OUT" THEME

Split/dual innervation appears 3 times across the limb — spot the pattern:

1) Brachialis (arm) — musculocutaneous + radial

2) Flexor digitorum profundus (forearm) — ulnar + anterior interosseous

3) Lumbricals (hand) — median (lateral 2) + ulnar (medial 2)

Recognizing this repeating structure is worth more than memorizing each fact in isolation.

CLINICAL QUICK-FIRE

Surgical neck fx → axillary n. → deltoid palsy, regimental badge numbness.

Mid-shaft humeral fx → radial n. → wrist drop.

Supracondylar fx (children) → brachial artery + median n.

Anterior shoulder dislocation → Bankart (IGL/labrum) + Hill-Sachs (humeral head) lesions.

Colles' (dorsal) vs Smith's (volar) distal radius fractures.
Scaphoid fx → snuffbox tenderness → AVN risk (retrograde supply).
Long thoracic n. → winged scapula. Accessory n. → trapezius palsy, shoulder droop.